Application for Exemption from Research Ethics Review

About this form:

This form is for researchers applying for ethics approval for human research projects that have been marked as exempt from review using the **ETHICS CHECKLIST FOR EXEMPT FROM REVIEW**.

The completed checklist must be attached as an attachment to this form

Attachments:

You need to include with this application the following supplementary documents

- A completed "exempt from review" checklist.
- A copy of the written permission to access the non-identifiable data set (if applicable).

Submitting the application form:

Submit your ***original*** and ***signed*** application form as well as any supplementary information to UMREC. Please get the application form signed by the Deputy Dean of Research.

You must not commence your study until you receive approval from UMREC

Please remove this cover page before submitting your application

Application for Exemption from Research Ethics Review

Application Number (office use)		**Approval Date** (office use)	

1 PROJECT TITLE	Self-Supervised Graph Neural Network for Network Intrusion Detection System.

Section 2: Researchers

2.1 PRINCIPAL INVESTIGATOR – STAFF RESEARCH / SUPERVISOR

Title	**First name, family name**	**phone**	**email**
Dr.	Muhammad Shahreeza Safiruz bin Kassim	+6018-3967974	shahreeza@um.edu.my

2.2 OTHER INVESTIGATOR/S

Title	**First name, family name**	**phone**	**email**

2.3 STUDENT PRINCIPAL INVESTIGATOR

Title	**First name, family name**	**phone**	**email**
Mr	*Adam Bin Razali*	+6017-4020027	22002146@siswa.um.edu.my

2.4 DEGREE/PROGRAMME:

☐Undergraduate (Bachelor of Computer Science - Artificial Intelligence, FSKTM) /	☐Master (Coursework)	☐Master (Mixed-mode)	☐PhD

2.5 RESEARCH GRANT DETAILS

Research funding/Grant	N/A (Self-funded / FYP Academic Research)
Amount of Research Grant	-

3. PROJECT DETAILS

3.1	**PROPOSED DURATION OF THE WHOLE RESEARCH PROJECT**	From:	March 2026	To:	Februrary 2027

3.2	**PROPOSED DURATION FOR THE DATA COLLECTION PHASE OF THE RESEARCH PROJECT**	From:	July 2026	To:	November 2026

3.3 IN PLAIN ENGLISH PROVIDE A SUMMARY OF THE RESEARCH PROJECT

See National Statement on Ethical Conduct in Human Research (2023), Section 1

Please state the aims and research objectives of the study, followed by a detailed explanation of the secondary data your study involves. Include how you will ensure the data remains anonymous and non-identifiable. Briefly describe the method of data analysis as well. (300-400 words)

This project aims to develop a Self-Supervised Graph Neural Network (GNN) framework to enhance Network Intrusion Detection Systems (NIDS) in detecting malicious network activity and abnormal traffic patterns.

The architecture of the proposed GNN solution has already been constructed and finalized. The primary objective of the current phase is to evaluate and test its model inference capabilities and performance when deployed on real-world, enterprise-scale network traffic.

The study relies strictly on secondary data analysis of existing, historical NetFlow network traffic telemetry records provided by Universiti Malaya. The dataset consists purely of network-layer metadata, including flow duration, source and destination port numbers, protocol types, packet counts, and byte counts. No individual payload content, browsing history, personal user data, or communication contents are collected, stored, or inspected.

To guarantee complete participant anonymity and data privacy, all Internet Protocol (IP) addresses within the network logs are fully masked and de-identified at source prior to research processing. The investigator cannot identify, re-identify, or trace any individual network user or host from the dataset. Data access is legally granted through appropriate institutional data-custodian authorizations strictly for academic research within Universiti Malaya.

The analytical methodology involves transforming the masked network telemetry into topological graph representations, where network nodes represent masked communication endpoints and edges denote flow connections. Self-supervised learning algorithms, including masked autoencoders and graph structure learning, will be benchmarked on the unlabeled network graphs to evaluate baseline anomaly detection accuracy and real-world inference speed without requiring manual labeling or compromising privacy.

4. DECLARATION BY RESEARCHERS

I/We accept responsibility for the conduct of the study in accordance with the principles of ethical research.

Adam Bin Razali		:	19/10/2026
Name	*(Signature of Principal Investigator)*		Date

Declaration by the Supervisor (if student is not the principal investigator)

I have informed the student of their responsibility to undertake this research in a manner that conforms with the principles of research ethics.

Dr. Muhammad Shahreeza Safiruz bin Kassim			19/10/2026
Name	*(Signature of Supervisor)*		Date

Declaration by the Deputy Dean of Research:

The project set out in the attached application, including the adequacy of its research design and compliance with recognised ethical standards, has the approval of the Faculty.

Name: Date:

(Signature of Deputy Dean of Research)

Faculty: Faculty of Computer Science & Information Technology (FSKTM) University Malaya